Simple Pendulum and Gravity

1. Definition

A simple pendulum consists of a small heavy bob suspended from a fixed support by a light, inextensible string. When displaced slightly from its mean position and released, it oscillates periodically.

The acceleration due to gravity (g) is the acceleration experienced by a freely falling body due to Earth's gravitational force.

2. Principle

For small angular oscillations, the time period of a simple pendulum depends on its length and the acceleration due to gravity.

The time period is given by:

$T = 2\pi\sqrt{(L/g)}$

Squaring both sides:

$T^2 = 4\pi^2L/g$

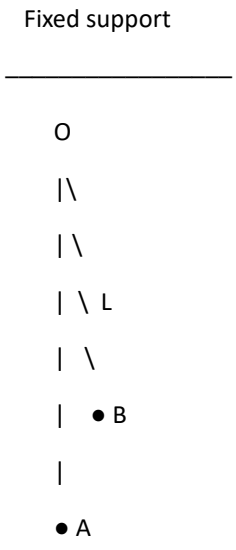
Therefore:

$g = 4\pi^2L/T^2$

Where:

- T = time period of oscillation
- L = effective length of the pendulum
- g = acceleration due to gravity

3. Diagram



A = Mean position

B = Displaced position

L = Length of pendulum

4. Apparatus Required

- Simple pendulum
- Stopwatch
- Metre scale
- Retort stand and clamp

5. Procedure

1. Suspend the bob from a fixed support using a light string.
2. Measure the length L from the point of suspension to the centre of the bob.
3. Displace the bob slightly and release it without pushing.
4. Measure the time t taken for 20 complete oscillations.
5. Calculate the time period using:

$T = t/20$

6. Calculate the value of g using:

$g = 4\pi^2L/T^2$

7. Repeat the experiment for different lengths and calculate the mean value of g.

6. Observation Table

Number of oscillations (n) = 20

S. No.	Length (L) in m	Time for 20 oscillations (t) in s	Time period (T = t/20) in s	T^2 (s^2)	$g = 4\pi^2L/T^2$ (m/s^2)
1	0.40	25.4	1.27	1.6129	9.79
2	0.60	31.1	1.555	2.4180	9.80
3	0.80	35.9	1.795	3.2220	9.80
4	1.00	40.1	2.005	4.0200	9.82

Sample Calculation

For the first observation:

Length, L = 0.40 m

Time for 20 oscillations, t = 25.4 s

Time period:

$T = t/20 = 25.4/20 = 1.27$ s

Square of time period:

$T^2 = (1.27)^2 = 1.6129\ s^2$

Acceleration due to gravity:

$g = 4\pi^2L/T^2$

$g = (4 \times 3.1416^2 \times 0.40)/1.6129$

$g \approx 9.79\ m/s^2$

7. Precautions

1. Keep the angular displacement small.
2. Measure the length up to the centre of the bob.
3. Release the bob gently without applying any force.
4. Avoid air currents and external disturbances.
5. Count complete oscillations carefully.

8. Result

The acceleration due to gravity is determined using the formula:

$g = 4\pi^2 L/T^2$

Mean acceleration due to gravity:

$g \approx 9.80\ m/s^2$

9. Conclusion

The time period of a simple pendulum increases with its length and decreases as the acceleration due to gravity increases. The experiment can be used to determine the local value of g.